
OCT Scan Disease Detector Project

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Link to Google Colab

Introduction

Optical coherence tomography (OCT) is one of the most widely performed retinal imaging procedures, and it is central to diagnosing several leading yet treatable causes of vision loss, including choroidal neovascularization (CNV), diabetic macular edema (DME), and drusen [Kermany et al., 2018a]. Detecting these conditions early allows for effective treatment, but reading the large volume of scans produced in practice depends on scarce specialist time [Choudhary et al., 2023]. The goal of this project is to train a convolutional neural network (CNN) that classifies a retinal OCT scan into one of four categories (CNV, DME, drusen, or Normal). To make the model both measurable and trustworthy, we compare it against a simple baseline and use Grad-CAM [Selvaraju et al., 2017] to check whether its predictions rely on clinically meaningful regions of the scan. Deep learning suits this task because the distinguishing signs are subtle and localized, such as small pockets of fluid or lipid deposits in specific retinal layers [Kermany et al., 2018a], and CNNs learn the relevant features directly from pixels. The model will be trained from scratch on the large public OCT dataset of Kermany et al. [2018a] and success will be judged by classification accuracy relative to the baseline, together with whether Grad-CAM highlights known disease features.

Illustration

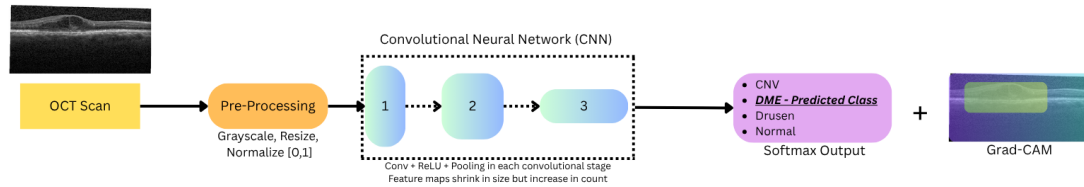


Figure 1: Illustration showing the overall project pipeline: Preprocessing, model evaluation, final prediction output, and a Grad-CAM overlay for analysis

Background & Related Work

Kermany et al. [2018a] introduced the public OCT dataset used in this project and showed that a transfer-learning model (InceptionV3) could classify the four retinal categories with expert-level accuracy; occlusion testing was used to highlight the regions behind each prediction. Transfer learning remains the dominant approach on this dataset; Choudhary et al. [2023] applied a VGG-19 model with ImageNet pretraining to the same four classes and reported high accuracy. The value of deep features over hand-designed ones has also been shown directly with this data. Nugroho [2018] compared handcrafted features (HOG and LBP) against deep networks and found the deep networks far more accurate, which motivates both our use of a CNN and our choice of a simple handcrafted-feature baseline. A separate but important concern is how the data is split, Tampu et al.

[2022] demonstrated that splitting OCT images without respecting patient boundaries inflates reported accuracy substantially, which directly informs our patient-level split. Finally, for interpretability, we use Grad-CAM [Selvaraju et al., 2017], a gradient-based method that highlights the image regions driving a CNN’s prediction without requiring any change to the network or retraining. Together, these works frame our approach, a CNN trained from scratch, compared against a classical baseline, and audited with Grad-CAM under a data split that avoids leakage.

Data Processing

The primary dataset is the public OCT collection of Kermany et al. [2018a], obtained from its Mendeley repository [Kermany et al., 2018b]. It is distributed as a training pool of approximately 108k labeled scans and a held-out test set of 1,000 images (250 per class), which we leave untouched for final evaluation. The classes are heavily imbalanced; Normal and CNV together exceed 80% of the training pool, while drusen makes up about 8% (Table 1, Appendix).

Each filename contains a patient identifier, which we use to split the training pool into training and validation at the patient level, roughly 90/10. Every image from a given patient is assigned to a single split, so no patient, and therefore no eye, appears in both. This prevents the inflated accuracy that occurs when near identical scans from one patient are separated across splits [Tampu et al., 2022]. Class proportions are approximately preserved across the two splits (Table 1, Appendix). Preprocessing is identical for every image, grayscale conversion, resizing to 224x224, and scaling the pixels to the [0,1] range. Figure 2 shows representative samples after this pipeline.

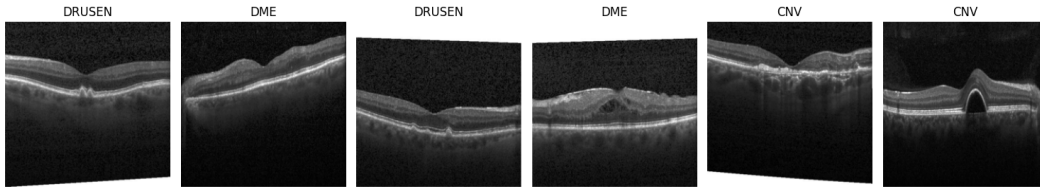


Figure 2: Samples of OCT scans after preprocessing steps.

The class imbalance is addressed during training with a class-weighted loss.

To assess generalization on unseen data, we additionally evaluate the final model on OCTDL [Kulyabin et al., 2024], an independent set acquired with a different scanner (Optovue Avanti) and patient population. We keep the three overlapping classes, Age-related macular degeneration (AMD), DME, and Normal, while discarding the rest. Because our CNV and drusen classes are both stages of AMD [Kermany et al., 2018a], we merge the model’s CNV and drusen predictions into a single AMD class so its outputs align with OCTDL’s labeling.

Architecture

The primary model is a convolutional neural network (CNN) trained from scratch, which suits the task because the features that distinguish the classes are small and localized. As shown in Figure 3, the network applies three convolutional blocks (each a 3 x 3 convolution, ReLU, and 2 x 2 max pooling) with the channel count increasing from 16 to 32 to 64, after which the feature maps are flattened and passed through a fully-connected layer of 128 units to a four-way output, giving roughly 6.45M parameters. It is trained with the Adam optimizer (learning rate 0.001) and a class-weighted cross-entropy loss to counter the class imbalance, with a batch size of 128 and early stopping (patience 3). Preprocessed images are decoded once and cached in memory, which allows training on the full training pool of 98,050 images rather than a subsample. The model reaches its lowest validation loss at epoch 5, at 85.2% validation accuracy, after which validation loss rises while training accuracy continues to climb, a clear overfitting pattern that early stopping halts (Table 2, Appendix).

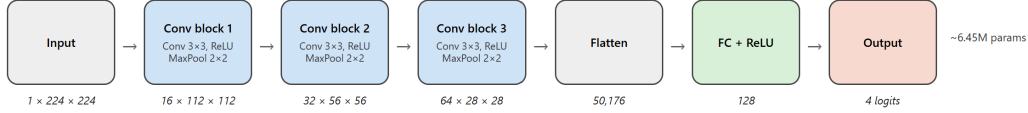


Figure 3: The primary CNN with three convolutional layers and two fully-connected layers

Baseline Model

As a baseline, a classical model that does not learn its own features is used, giving a floor against which the CNN is measured and isolating how much its learned features contribute. Histogram of Oriented Gradients (HOG) feature vectors [Dalal and Triggs, 2005] are extracted from each preprocessed scan and used to train a support vector machine (SVM) with a radial basis function (RBF) kernel [Cortes and Vapnik, 1995]; HOG is a standard, non-learning descriptor of local edge structure and the SVM is a strong classifier that is simple to implement and requires little tuning, making the pair a reasonable baseline. Both stages are illustrated in Figure 5 (Appendix). Because SVM training scales poorly with dataset size [Tsang et al., 2005], the model is trained on a balanced subsample of 1,500 images per class and evaluated on the same held-out 1,000-image test set used to evaluate the finalized CNN, reaching 75.5% accuracy. The confusion matrix (Table 3, Appendix) shows drusen as the weakest class, most often misclassified as CNV, indicating HOG cannot reliably distinguish drusen from CNV, the early and advanced versions of the same disease. The main challenge is the scaling limitation, which forces the use of a subsample and motivates the CNN, which learns features directly and can scale to the full dataset.

Quantitative Results

On the held-out 1,000 image test set, the CNN reaches 85.2% accuracy, well above the baseline’s 75.5%, confirming that learned features outperform the handcrafted HOG descriptor. Test accuracy matches the 85.2% validation figure, indicating the model generalizes to the held-out set. Per class performance is uneven; the model is strongest on CNV and DME (recall of 0.95 and 0.91) but weakest on drusen, which has a recall score of 0.65, most often being misclassified as CNV (Table 4, Appendix). This mirrors the baseline’s failure mode and reflects the clinical similarity of drusen and CNV as the early and advanced stages of the same disease.

Qualitative Results

To assess whether predictions are driven by meaningful image regions, we apply Grad-CAM [Selvaraju et al., 2017] to the trained model on test scans (Figure 4). On correctly classified DME and Normal scans, attention often falls on plausible retinal structure such as fluid pockets and the retinal layers. However, attention is inconsistent on CNV; even across correctly classified CNV scans, the highlighted regions vary widely and frequently drift to the edge or background of the images. The drusen scans misclassified as CNV show a related pattern, with attention on upper and lower retinal regions rather than the deep retinal pigment epithelium and Bruch’s membrane, where drusen deposits actually manifest [Kulyabin et al., 2024]. This is consistent across the wider set of examples in Figure 6 (Appendix).

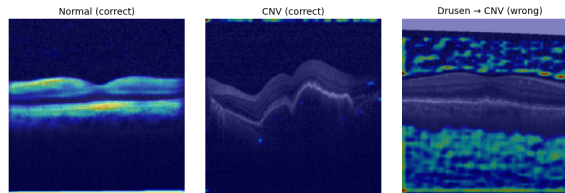


Figure 4: A correctly classified Normal scan, a correctly classified CNV scan with attention that does not localize to a clear lesion, and a drusen scan misclassified as CNV.

Evaluate on New Data

To test generalization beyond the training distribution, we evaluate the finalized model on OCTDL [Kulyabin et al., 2024], an independent set of 1,710 scans acquired with a different scanner and patient population and used in no way during training or tuning. Since OCTDL groups drusen and neovascular cases under a single AMD label, we merge the model’s CNV and drusen predictions into AMD, giving a three-class problem (AMD, DME, Normal). The model reaches 74.0% accuracy, down from 85.2% on the held-out test set, a drop expected with the changes introduced by the new scanner. Performance is uneven across classes (Table 5, Appendix). AMD is recognized reliably (recall 0.87), but Normal scans are frequently misclassified as DME (recall 0.34). The AMD category transferring well while Normal does not suggests that the model captures the broad AMD category well, though not necessarily the finer drusen-CNV distinction it struggles with elsewhere, while also relying on scanner specific information that does not carry over, an unsurprising limitation for a model trained from scratch on a single source.

Discussion

The model performs well for a from-scratch CNN, 85.2% on the held-out test set is comfortably above the 75.5% baseline, confirming that learned features outperform handcrafted HOG descriptors for OCT classification. The clearest weakness is drusen, with a recall score of 0.65 and most often confused with CNV. This is a clinically grounded error as drusen and CNV are the early and advanced stages of the same disease, so their OCT signatures overlap, and the same confusion appears in the baseline. The per-epoch change shows an overfitting pattern, validation loss bottoming at epoch 5 while training accuracy climbs past 95%, which the class-weighted loss and early stopping help mitigate.

The external evaluation is the most informative result. Accuracy falls from 85.2% to 74.0% on OCTDL; the AMD category transfers well, but Normal scans are frequently read as DME, suggesting the model has learned genuine disease structure alongside scanner specific texture that does not survive a change of hardware. Grad-CAM reinforces this reading. Attention to lesions is inconsistent and on misclassified drusen it falls on the retinal boundaries rather than the deposits, so while the model reaches useful accuracy, its predictions are not driven by consistently interpretable features. The main lesson is that although the model performs strongly on its source dataset, training on a single source from one scanner limits its real-world readiness.

Ethical Considerations

The training data raises a few ethical concerns. It was collected from a small number of clinics, acquired on a single OCT vendor’s system [Choudhary et al., 2023], and only from adult patients [Kermany et al., 2018a], so the model may not generalize to other imaging devices or population demographics, a risk demonstrated by the accuracy drop on the OCTDL scanner. The classes are also imbalanced, with conditions such as drusen far less represented, so the model may learn the rarer diseases less reliably even though they remain clinically important. Deployment carries real consequences, a false negative could delay referral for a condition that can cause irreversible vision loss [Kermany et al., 2018a], so errors carry real clinical consequences. Grad-CAM partially mitigates this by making each prediction inspectable, allowing a clinician to check whether the model attended to clinically meaningful regions rather than trusting it blindly.

References

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A Appendix

Table 1: OCT images per class in each split. The training pool is split at the patient level ($\sim 90/10$); the supplied test set is held out.

Class	Train	Validation	Test
CNV	33,932	3,273	250
DME	10,227	1,121	250
Drusen	7,863	753	250
Normal	46,028	5,112	250
Total	98,050	10,259	1,000

Table 2: Per-epoch metrics for the primary CNN trained on the full dataset. The highlighted row (epoch 5) has the lowest validation loss; early stopping (patience 3) halts training after epoch 8.

Epoch	Train loss	Train acc	Val loss	Val acc
1	1.037	59.4%	0.672	74.9%
2	0.607	80.3%	0.695	72.7%
3	0.467	84.8%	0.503	81.6%
4	0.379	87.6%	0.544	81.2%
5	0.302	89.9%	0.428	85.2%
6	0.233	92.0%	0.466	85.1%
7	0.173	94.0%	0.464	85.0%
8	0.130	95.3%	0.577	83.4%

Table 3: Baseline confusion matrix on the 1,000-image test set (rows: true class, columns: predicted). Overall test accuracy 75.5%.

True \ Pred	CNV	DME	Drusen	Normal
CNV	229	19	2	0
DME	30	214	2	4
Drusen	84	20	122	24
Normal	4	31	25	190

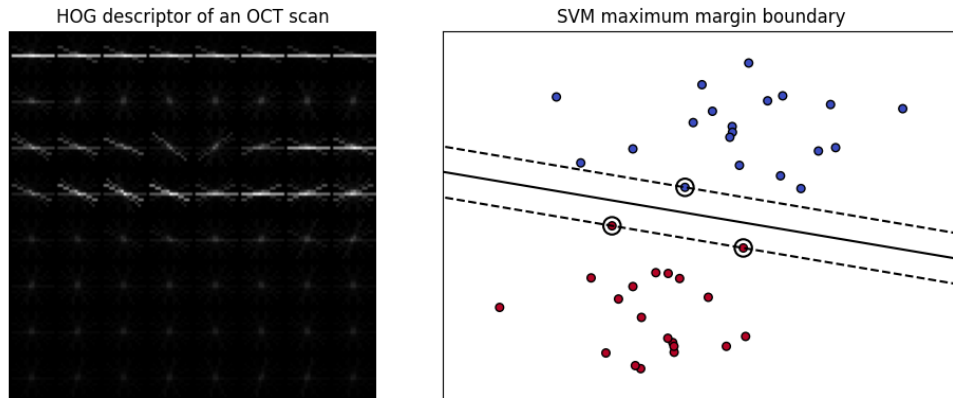


Figure 5: The two stages of the baseline. Left: a visualization of the HOG descriptor of an OCT scan. Right: an illustration of the SVM boundary separating two classes, with support vectors circled.

Table 4: CNN confusion matrix on the 1,000-image test set (rows: true class, columns: predicted). Overall test accuracy 85.2%.

True \ Pred	CNV	DME	Drusen	Normal
CNV	237	11	1	1
DME	15	227	1	7
Drusen	54	11	162	23
Normal	2	12	10	226

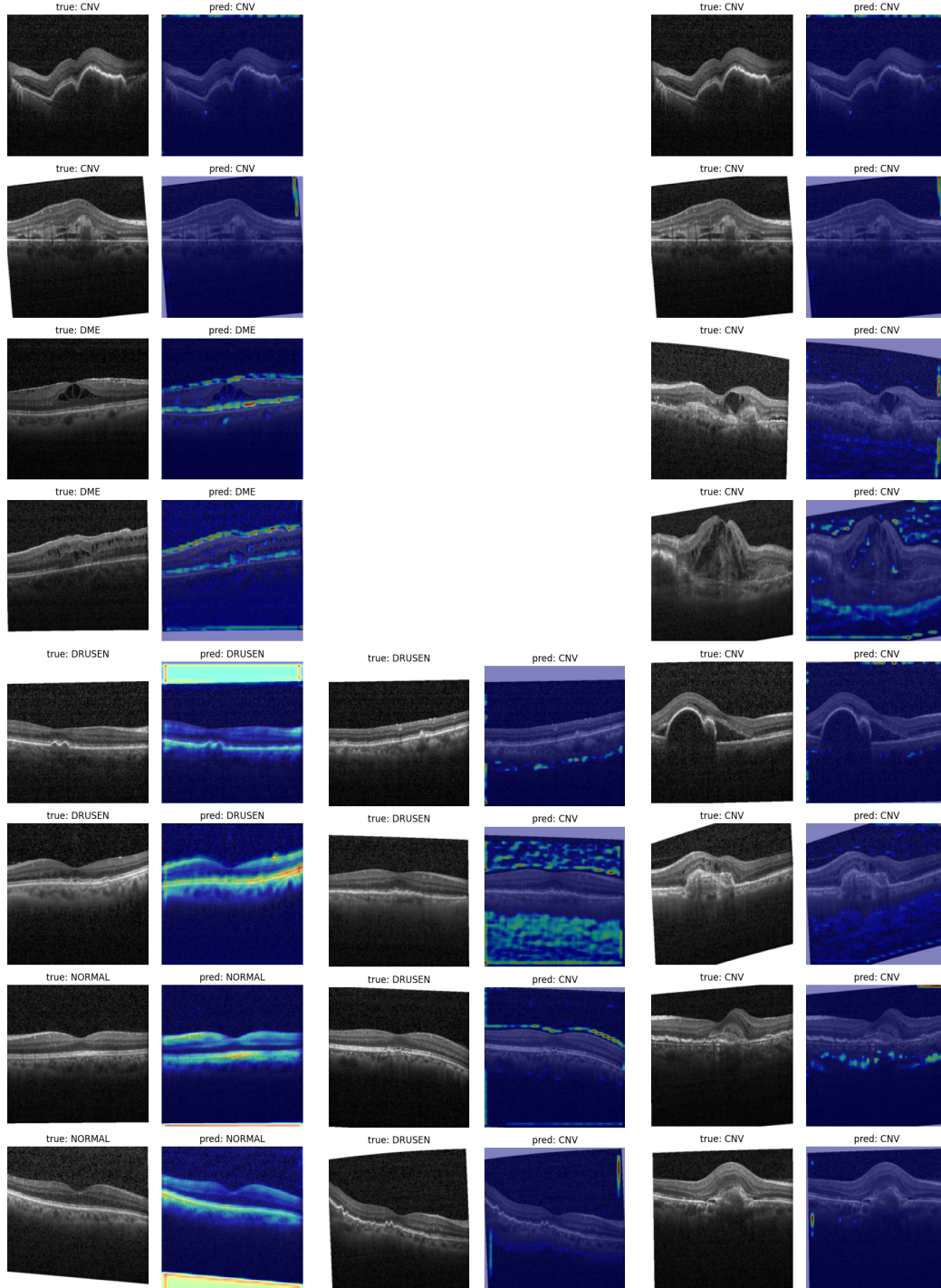


Figure 6: Grad-CAM overlays. Left: correctly classified scans across classes. Center: drusen scans misclassified as CNV. Right: correctly classified CNV scans, showing inconsistent attention.

Table 5: OCTDL confusion matrix (rows: true class, columns: predicted). Overall accuracy 74.0% on 1,710 scans.

True \ Pred	AMD	DME	Normal
AMD	1074	136	21
DME	65	79	3
Normal	29	190	113